Sahaj Bhadja

(213) 284-4651 sahaj.bhadja@gmail.com sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years of experience building **production APIs**, **automation systems**, cross-platform services, and fault-tolerant workflows spanning cloud applications, simulation, and developer-facing integrations.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Tripled concurrent throughput across 1K+ jobs by restructuring **fault-tolerant API services** with explicit state transitions, failure isolation, recoverable background processing, and observable workflows under sustained production load reliably.
- Achieved 99%+ reliability across 5K+ monthly jobs through **automation pipelines**, implementing validation, reconciliation, error handling, cloud storage, CI/CD releases, and stable production monitoring workflows reliably at scale.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive user workflows during concurrent processing and sustained production traffic reliably.

USC AD Labs | *ML Research Assistant* | *Los Angeles, CA* **January 2025 – May 2025**
- Improved long-horizon trajectory planning by 18% in simulation by building a **Decision Transformer pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and structured evaluation workflows reliably.
- Improved closed-loop planning stability by 15% through a **spatial-temporal encoder** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across complex simulation scenarios.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) | *India*
- Synchronized 2M+ records through an eight-service **API and automation platform** covering authentication, resources, customers, billing, leads, permissions, and notifications with Node.js, WebSockets, and Azure SQL across locations.
- Reduced resource wastage by 42% through **automated testing and deployment**, implementing role-based access, WebSocket updates, Azure monitoring, Docker/Jenkins pipelines, and ARM templates across eight modular service domains.

Skills

Languages: Python, C++, Java, TypeScript, JavaScript, SQL, Bash
APIs & Automation: REST APIs, API Design, SDK Design, Microservices, Background Workers, Asynchronous Processing, WebSockets, Authentication, Error Contracts, Workflow Automation
Systems: Object-Oriented Design, Data Structures, Algorithms, Concurrency, Multithreading, Fault Tolerance, Failure Isolation, State Management, Caching, Networking
Cloud & Data: AWS, Azure, Docker, Kubernetes, Linux, PostgreSQL, MySQL, MongoDB, AWS RDS, Azure SQL
Quality & Delivery: Unit Testing, Integration Testing, Automated Testing, PyTest, Mockito, Jest, Git, GitHub, Jenkins, CI/CD, Debugging, Monitoring, Observability, Technical Documentation

Projects

Print Workflow Automation SDK - Design Concept
- Designed a **print-workflow automation SDK** concept exposing Python clients for job preparation, validation, queue submission, long-running operation polling, stable error contracts, authentication, retries, and developer-focused documentation examples.

Device Fleet Telemetry Gateway - Design Concept
- Designed a **device-fleet telemetry gateway** concept collecting printer status, consumables, job history, and progress through authenticated APIs, with asynchronous processing, network-failure recovery, monitoring, and tenant-isolated operational dashboards.

CPU-Scheduling Simulator | Live
- Built a **CPU-scheduling simulator** implementing FCFS, SJF, SRTF, Priority, and Round Robin algorithms with configurable workloads, timeline visualization, comparative metrics, and deterministic validation across scheduling scenarios reliably.